

RESEARCH ARTICLE

Solid-State-Enabled Three-Stage Delithiation Pathway of Li_5FeO_4 for Silicon-Based All-Solid-State Batteries

Tingting Liu^{1,2,3} | Shengjie Xia^{1,2,4} | Kaiyong Tuo^{1,2,3} | Jiamin Fu^{1,2} | Junwu Sang^{1,2} | Suze Liang^{1,2,3}  | Zhimin Zhou^{1,2,4} | Mengfei Zhu^{1,2,3} | Mingfeng Wei^{1,2,3}  | Ziqing Wang^{1,2,3}  | Luting Xie^{1,2,4} | Hongrui Huang^{1,2,4} | Jian Peng^{1,2}  | Qiaobao Zhang⁵  | Xueliang Sun^{1,2}  | Changhong Wang^{1,2} 

¹Eastern Institute for Advanced Study, Ningbo Institute of Digital Twin, Eastern Institute of Technology, Ningbo, China | ²Zhejiang Key Laboratory of All-Solid-State Battery, Ningbo Key Laboratory of All-Solid-State Battery, Ningbo, China | ³University of Science and Technology of China, Hefei, China | ⁴Shanghai Jiao Tong University, Shanghai, China | ⁵State Key Laboratory of Physical Chemistry of Solid Surfaces, College of Materials, Xiamen University, Xiamen, China

Correspondence: Qiaobao Zhang (zhangqiaobao@xmu.edu.cn) | Xueliang Sun (xsun@eitech.edu.cn) | Changhong Wang (cwang@eitech.edu.cn)

Received: 26 June 2026 | **Revised:** 8 September 2026 | **Accepted:** 10 September 2026

Keywords: all-solid-state batteries | Li_5FeO_4 | prelithiation additive | silicon anode

ABSTRACT

Li_5FeO_4 is a prelithiation additive for compensating irreversible lithium loss in lithium-ion batteries, yet its function and mechanism in all-solid-state batteries (ASSBs) remain largely unexplored. Here, we demonstrate that nanosized Li_5FeO_4 serves as an effective cathode prelithiation additive in halide-based ASSBs. Through mechanochemical milling, pristine micrometer-sized Li_5FeO_4 is reduced to approximately 500 nm, leading to a significant enhancement in delithiation capacity from 43.1 to 752.5 mAh g⁻¹. This enhancement arises from nanosizing-induced local structural disorder and improved electronic conductivity. Mechanistic analyses reveal a unique three-stage delithiation pathway in the all-solid-state reaction environment, involving low-potential lattice oxygen oxidation, coupled oxygen/iron oxidation, and lattice oxygen oxidation accompanied by iron reduction. Guided by this mechanism, nanosized Li_5FeO_4 replenishes lithium consumed by silicon anodes, enabling room-temperature silicon-based ASSBs with 70% capacity retention over 1000 cycles and stable operation from -10°C to 55°C. Moreover, a silicon-based all-solid-state pouch cell delivers an energy density of 473.3 Wh kg⁻¹ (calculated on electrode mass) and retains 86.1% capacity after 500 cycles under 24.5 MPa. This work reveals the distinct delithiation pathway of Li_5FeO_4 in an all-solid-state reaction environment, thus providing mechanistic guidance for developing efficient prelithiation strategies for high-energy-density ASSBs.

1 | Introduction

All-solid-state batteries (ASSBs) are widely regarded as promising next-generation energy-storage systems because of their potential to combine high safety, high energy density, and broad operating-temperature tolerance [1–6]. By replacing flammable liquid electrolytes with inorganic solid electrolytes, ASSBs offer opportunities to pair high-capacity electrodes, such as lithium metal or silicon anodes, with high-voltage cathodes. However, the practical energy output of ASSBs is still strongly constrained by irreversible lithium loss during the initial cycles. This lithium

inventory loss mainly arises from interphase formation, interfacial side reactions, and irreversible reactions at high-capacity anodes, which lower the initial Coulombic efficiency and reduce the amount of cyclable lithium available for long-term operation [7–15]. The issue is particularly severe for silicon-based ASSBs, where the large surface area, substantial volume change, and continuous interfacial reconstruction of silicon can consume considerable active lithium during early cycling.

Prelithiation is an effective strategy to compensate for irreversible lithium loss by introducing additional lithium sources into

the battery. Existing prelithiation approaches can generally be divided into anode prelithiation [16–19] and cathode prelithiation [20–24]. Anode prelithiation, including chemical, electrochemical, and direct-contact lithiation, can provide lithium compensation but often suffers from uneven lithium distribution, poor process compatibility, environmental sensitivity, and safety concerns. These issues become even more challenging in ASSBs because solid–solid interfaces are mechanically rigid and chemically sensitive. In contrast, cathode prelithiation using lithium-rich sacrificial additives provides a simpler and more practical route. Lithium-rich compounds such as Li_2O [21, 25, 26], Li_2O_2 [27, 28], Li_6CoO_4 [29–31], and Li_5FeO_4 [32–38] can be incorporated into cathode composites and release lithium during the first charge, thereby compensating lithium consumed at the anode without directly disturbing the anode interface. This feature makes cathode prelithiation especially attractive for ASSBs, where interfacial compatibility and processing simplicity are critical for practical cell fabrication.

Among various cathode prelithiation additives, Li_5FeO_4 (LFO) is particularly appealing because of its high theoretical delithiation capacity of 867 mAh g^{-1} , low cost, and compatibility with the operating voltage range of commercial cathodes such as LiCoO_2 and Ni-rich layered oxides [39, 40]. Structurally, Li_5FeO_4 possesses an antifluorite framework rich in Li–O–Li configurations, which can create high-energy oxygen states and enable oxygen-redox-assisted lithium release beyond conventional transition-metal redox. The participation of both Fe and O redox provides the possibility of multielectron charge compensation and high capacity. However, oxygen redox also brings intrinsic complexity. In liquid-electrolyte batteries, oxidized oxygen species are often associated with irreversible oxygen release, structural degradation, and parasitic reactions with electrolyte components. Although Li_5FeO_4 has been widely investigated as a prelithiation additive in conventional liquid-electrolyte lithium-ion batteries, its effectiveness and delithiation mechanism in ASSBs remains largely unexplored.

In this work, we demonstrate that nanosized Li_5FeO_4 can serve as an effective cathode prelithiation additive for halide-based ASSBs. Through mechanochemical milling, pristine micrometer-sized Li_5FeO_4 is reduced to approximately 500 nm, leading to a dramatic increase in delithiation capacity from 43.1 to 752.5 mAh g^{-1} . Mechanochemical milling induces structural disorder, changes the local Fe coordination environment, and improves electronic conductivity, thereby enabling deeper delithiation. Combined mechanistic investigations further uncover a unique three-stage delithiation pathway in the all-solid-state reaction environment, involving low-potential lattice oxygen oxidation, coupled oxygen/iron oxidation, and lattice oxygen oxidation accompanied by iron reduction. Guided by this mechanism, nanosized Li_5FeO_4 is further incorporated into cathode composites to compensate lithium loss in Si-based ASSBs. The released lithium effectively replenishes lithium consumed by silicon anodes during initial cycling, enabling room-temperature Si-based ASSBs with 70% capacity retention over 1000 cycles and stable operation from -10°C to 55°C . Moreover, a Si-based all-solid-state pouch cell delivers an energy density of 473.3 Wh kg^{-1} (based on electrode mass) and retains 86.1% capacity after 500 cycles under 24.5 MPa. This work reveals the distinct delithiation chemistry of Li_5FeO_4 in an all-solid-state reaction environment

and establishes nanosized Li_5FeO_4 as an effective prelithiation additive for developing high-energy-density ASSBs.

2 | Results and Discussion

2.1 | Structural Optimization of LFO Toward Enhanced Electrochemical Performance in ASSBs

To investigate the electrochemical behavior of the LFO in ASSBs, pristine micrometer-sized LFO was first evaluated as a cathode material in three different cell configurations: a conventional liquid electrolyte battery using 1 M LiPF_6 in EC/DMC, a sulfide-based ASSB using $\text{Li}_6\text{PS}_5\text{Cl}$ (LPSC), and a halide-based ASSB using Li_3InCl_6 (LIC) (Figures S1 and S2). Consistent with previous reports [35], the liquid electrolyte battery delivers a charge capacity of 740 mAh g^{-1} when charged to 4.7 V. In sharp contrast, the two ASSBs exhibit significantly lower capacities, delivering only 44 mAh g^{-1} in both LIC- and LPSC-based cells. Interestingly, after mechanochemical milling, the resulting LFO exhibits a remarkable increase in charge capacity from 43.1 to 752.5 mAh g^{-1} in the LIC-based ASSB and from 44.6 to 365.4 mAh g^{-1} in the LPSC-based ASSB (Figure 1a,b). By contrast, the capacity of ball-milled LFO in the liquid-electrolyte cell remains largely unchanged (Figure 1c). Moreover, the charge–discharge profiles reveal that ball-milled LFO undergoes three distinct electrochemical stages in both halide- and sulfide-based ASSBs, whereas only two stages are observed in the liquid-electrolyte cell. This difference suggests that the solid-state reaction environment fundamentally alters the delithiation pathway of LFO. The underlying mechanism responsible for this solid-state-specific electrochemical behavior is discussed in the following sections.

Multiple characterization techniques were employed to elucidate the changes in the physicochemical properties of LFO induced by mechanical milling. Scanning electron microscope (SEM) reveals that mechanical milling reduces the particle size of LFO from $\sim 8 \mu\text{m}$ to $\sim 500 \text{ nm}$ in Figure S3. Consistently, peak broadening and intensity reduction in both x-ray diffraction (XRD, Figure S4) and synchrotron-based x-ray diffraction (sXRD, Figure 1d) patterns, together with the disappearance of Raman peaks (Figure S5), indicate a significant reduction in the crystallinity of LFO-BM. Quantitatively, the averaged apparent crystallite size decreases from 50.7 ± 10.0 to $13.3 \pm 3.6 \text{ nm}$ ($\sim 74\%$ reduction), while the relative crystalline diffraction contribution decreases to only 24.5% of that of pristine LFO, confirming that mechanical milling substantially disrupts the long-range crystallographic order. Consistent with the diffraction results, transmission electron microscopy (TEM) further reveals the local structural evolution (Figure 1e). The well-defined (222) lattice fringes observed in pristine LFO are replaced by a highly disordered structure after milling, indicating that mechanical milling introduces substantial local structural disorder.

Figure 1f shows Fe K-edge x-ray absorption near-edge structure (XANES) spectra of LFO and LFO-BM. The pre-edge peak A arises from the mixing of Fe 3d and 4p orbitals, while shoulder peak B originates from the overlap between Fe 4p and O 2p orbitals, both signatures of tetrahedrally coordinated FeO_4 units [33, 35, 41]. LFO-BM sample exhibits a reduced intensity of peak A and the complete disappearance of peak B, which directly

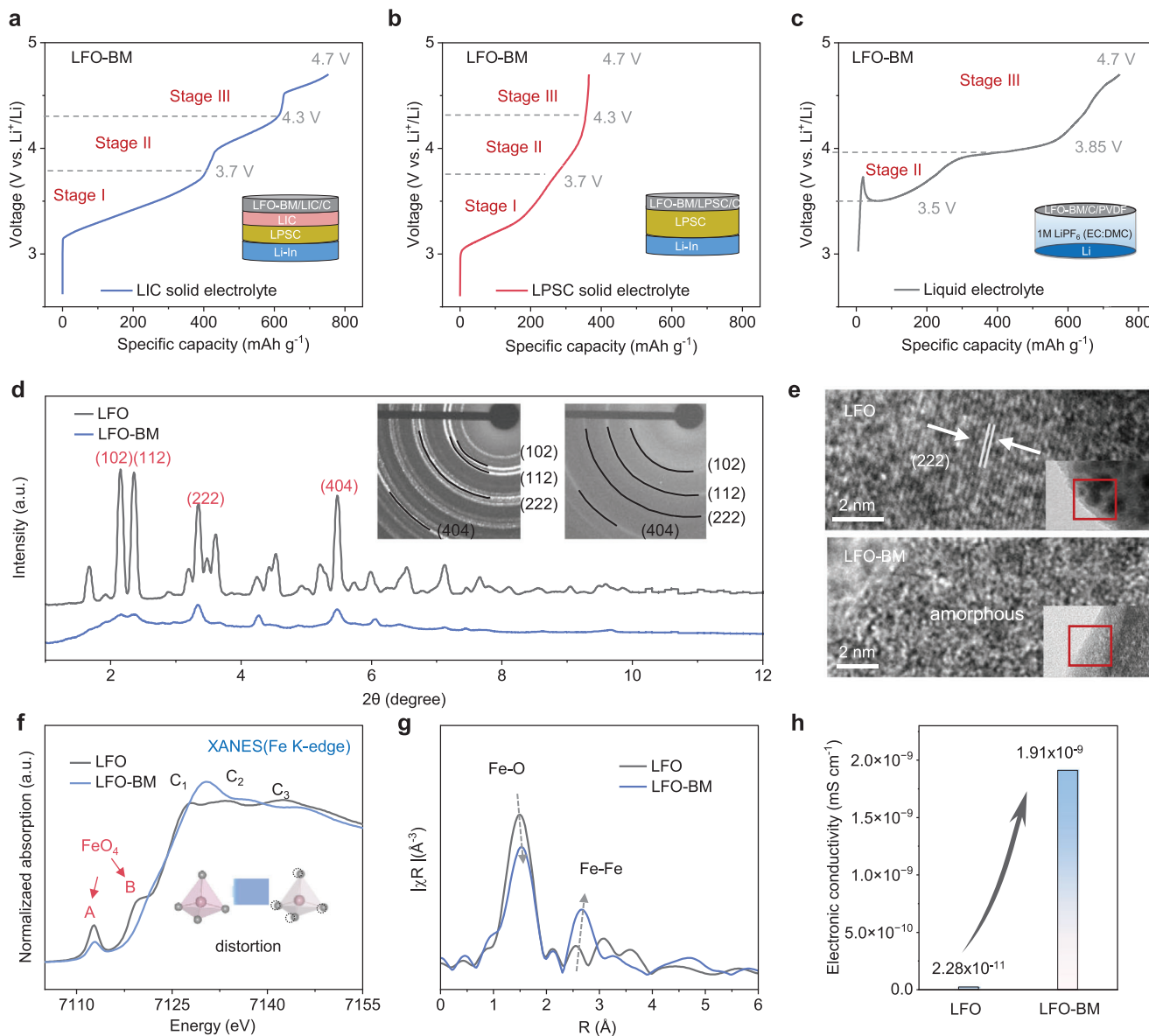


FIGURE 1 | Electrochemical behavior and local structure analysis of LFO-BM.

reflects the distortion and partial breakdown of the tetrahedral FeO_4 coordination environment. These spectral changes indicate substantial reconstruction of the local Fe coordination environment induced by mechanical milling [34]. The main peaks C_1 – C_3 arise from dipole-allowed transitions from the Fe 1s core level to unoccupied Fe 4p states, with their spectral profiles being sensitive to medium-range order and coordination environments beyond the first shell. Thus, the enhancement of peaks C_1 – C_3 results from the disruption of coordination symmetry.

The corresponding k^2 -weighted EXAFS oscillations before Fourier transformation are provided in Figure S6. The observed changes in oscillation amplitude and phase are consistent with the Fourier-transformed Fe K-edge extended x-ray absorption fine structure (FT-EXAFS) results discussed below. The FT-EXAFS spectra further reveal changes in the local Fe coordination environment (Figure 1g). The first peak at ~ 1.5 Å is attributed to Fe–O scattering contribution associated with the tetrahedral

FeO_4 coordination in the antifluorite structure. After mechanical milling, this peak becomes noticeably weaker and broader, indicating increased local structural disorder and distortion of the Fe–O coordination environment [36, 39]. Meanwhile, the Fe–Fe related scattering contribution at an apparent radial distance of ~ 2.6 Å becomes more distinguishable. Since a weak Fe–Fe related feature is already present in pristine LFO, this observation suggests the preservation of intrinsic short-range Fe–Fe structural correlations rather than the formation of new Fe–Fe bonds. Additionally, the Fe L-edge and O K-edge spectra (Figures S7 and S8) exhibit no distinct energy shifts while showing clear peak broadening. Therefore, these results indicate that mechanical milling significantly disrupts the long-range crystallographic order of LFO while reconstructing the local Fe coordination environment. Meanwhile, the enhanced electronic conductivity of LFO-BM is attributed to the combined effects of particle-size refinement, structural disorder, and defect generation induced by mechanical milling, as shown in Figure 1h and Figure S9.

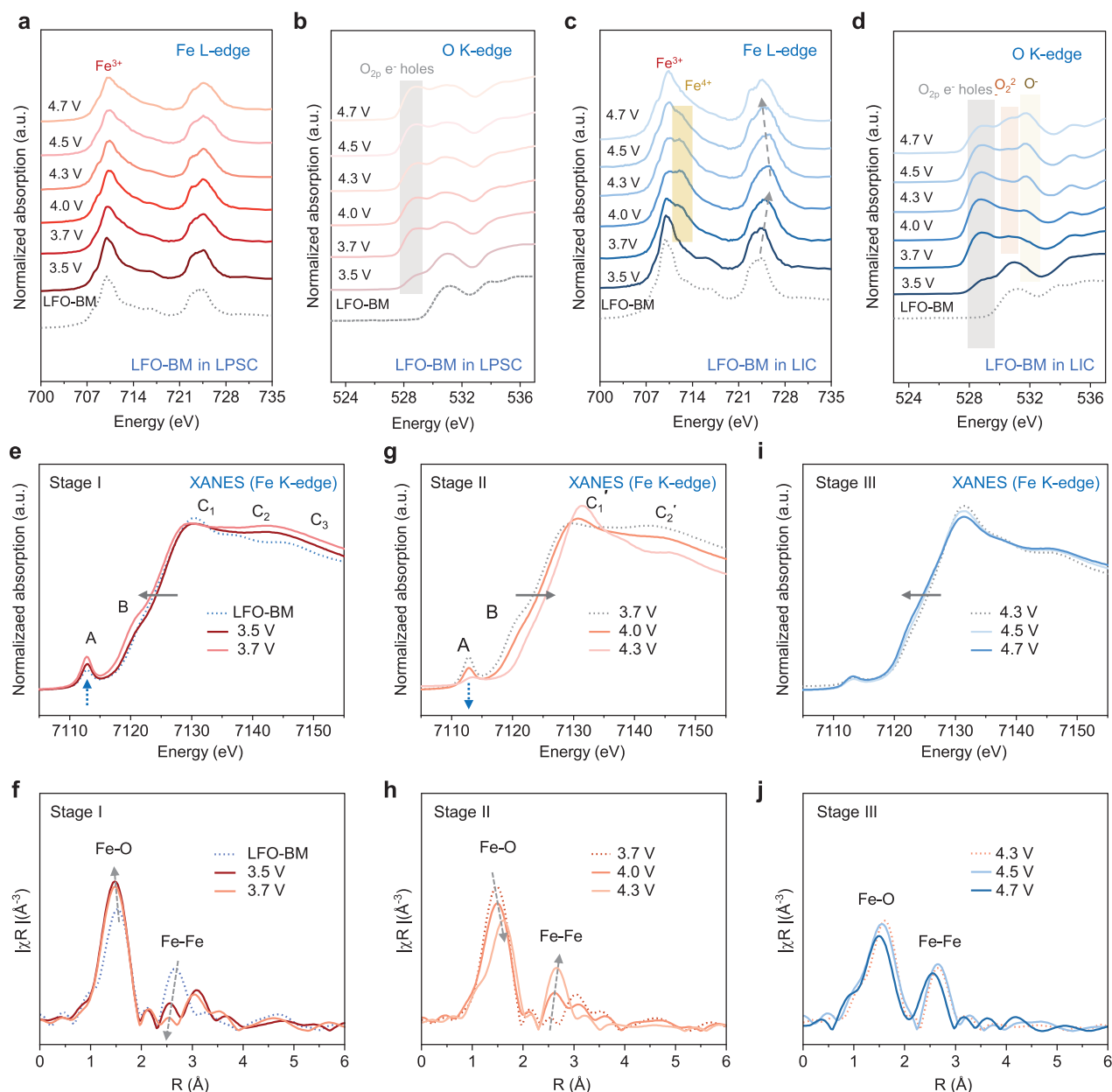


FIGURE 2 | Solid-State-Enabled Delithiation Mechanism of LFO-BM in ASSBs.

2.2 | Three-Stage Redox Mechanism of LFO-BM in ASSBs

Figure 2a,b present the ex situ Fe L-edge and O K-edge soft x-ray absorption spectroscopy (sXAS) of LFO-BM within a sulfide-based (LPSC) ASSB during the initial charge to 4.7 V. The Fe valence state shows no significant shift, while the O K-edge exhibits a progressively enhanced O 2p hole-state feature at ~528 eV [39, 42, 43], confirming oxygen as the primary redox-active species throughout Stage I, Stage II, and Stage III (Figure S10). Notably, the oxidized lattice oxygen species generated during anionic redox progressively oxidize the LPSC electrolyte, triggering its severe oxidative decomposition. The formation of S–O and P–O bonds was detected by the ex-situ x-ray photoelectron spectroscopy (XPS) as low as 3.7 V (Figure S11) [44–47], and these signals intensified markedly upon charging to 4.7 V, indicating that lattice oxygen oxidation leads to LPSC

electrolyte functional failure. Consequently, LFO-BM delivers a lower capacity in LPSC ASSBs than that achieved in LIC ASSBs.

Galvanostatic charge profiles of LFO-BM in various electrolyte systems: (a) halide-based (LIC), (b) sulfide-based (LPSC), and (c) conventional liquid electrolyte (1 M LiPF₆ in EC/DMC), all tested at 0.02 C (1 C = 867 mAh g^{−1}) with a 4.7 V cutoff voltage. (d) sXRD patterns of LFO and LFO-BM with the inset showing the corresponding structural models. (e) The TEM images of LFO and LFO-BM. (f) Fe K-edge XANES spectroscopy of LFO and LFO-BM. (g) FT-EXAFS spectra in R-space of LFO and LFO-BM, elucidating the local coordination environment. (h) Comparison of electronic conductivity between LFO and LFO-BM.

LFO-BM underwent a multistep redox reaction in the LIC ASSBs, enabled by the high-voltage stability and enhanced oxidative stability of the LIC electrolyte (Figure S12). The reaction process

evolves in three consecutive stages, corresponding to stage I (lattice oxygen oxidation), stage II (lattice oxygen and iron co-oxidation) and stage III (lattice oxygen oxidation accompanied by iron reduction).

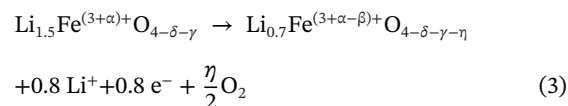
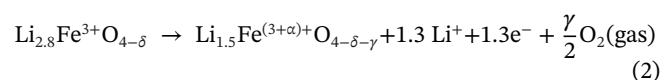
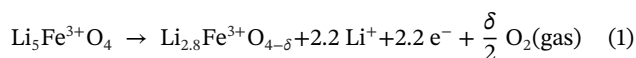
In stage I (up to ~3.7 V), the ex situ Fe L-edge and O K-edge sXAS spectra (Figure 2c,d), in agreement with the behavior in LPSC ASSBs, show clear lattice oxygen oxidation with no change in the Fe valence state, indicating that charge compensation is dominated by lattice oxygen redox. The structure evolution of LFO-BM electrode is traced by the ex situ Fe K-edge XAS in Figure 2e,f. The corresponding k²-weighted EXAFS oscillations before Fourier transformation are shown in Figure S13 and display systematic changes in both oscillation amplitude and phase throughout Stage I-III, consistent with the FT-EXAFS results discussed below. The gradual intensification of pre-edge peaks A and B (FeO₄ tetrahedral coordination) is accompanied by a strengthening Fe–O signal and the disappearance of Fe–Fe scattering in the FT-EXAFS (Figure 2f), demonstrating that the lattice oxygen oxidation results in structural rearrangement.

In stage II (3.7–4.3 V), Fe³⁺ oxidizes to Fe⁴⁺, accompanied by enhanced electron hole and oxidized oxygen species (Figure 2c,d), demonstrating coupled oxygen/iron oxidation. This triggers a coordination transition from tetrahedral FeO₄ to disordered octahedral FeO₆, as confirmed by the spectral evolution in Figure 2g: the weakening of peak A, disappearance of peak B, and transformation of peak C₁ into C₁'. This transition is also reflected in the FT-EXAFS by a broadened Fe–O scattering path and the emergence of a Fe–Fe contribution, indicating the formation of a disordered rocksalt-type framework.

In stage III (4.3 ~ 4.7 V), Fe⁴⁺ reduces back to Fe³⁺, coupled with a continued rise in oxidized oxygen species (Figure 2c,d). Meanwhile, the Fe K-edge shifts to lower energy in Figure 2g, while its FT-EXAFS profile shows no major change in the local Fe–O and Fe–Fe coordination (Figure 2j). Together with the persistent O₂ release detected by differential electrochemical mass spectrometry (DEMS) during charging (Figure S14), these findings conclusively demonstrate a charge-compensation mechanism dominated by concurrent iron reduction and lattice oxygen oxidation.

(a, b) Fe L-edge and O K-edge sXAS spectra of LFO at various states of charge (SOCs) in the LPSC electrolyte. (c, d) Fe L-edge and O K-edge sXAS spectra of LFO at different SoCs in the LIC electrolyte. (e–g) Fe K-edge XANES spectra of LFO-BM at different SoCs in the LIC electrolyte, highlighting the electronic structure evolution. (h–j) FT-EXAFS spectra in R-space and the corresponding structural parameters for Stages I, II, and III, respectively.

Therefore, based on the measured capacity, the extraction of approximately 4.3 Li⁺ can be expressed as follows:



Stage I (low-potential lattice oxygen oxidation), Stage II (oxygen and Fe co-oxidation), and Stage III (lattice oxygen oxidation accompanied by iron reduction). Notably, the intrinsic Stage I plateau is suppressed in liquid electrolytes (1 M LiPF₆ in EC/DMC) due to side reactions and sluggish interfacial kinetics. (b) Schematic comparing the structural transitions and redox pathways of LFO-BM between solid-state and liquid-phase environments.

The complete electrochemical evolution of LFO-BM in ASSBs is schematically illustrated in Figure 3a, with the corresponding structural evolution depicted in Figure 3b. The redox behavior in Stage II and Stage III is consistent with the reported delithiation mechanism of LFO in liquid electrolytes [25, 39, 47]. However, compared with its behavior in liquid systems, LFO-BM in the solid-state environment exhibits an additional charge compensation process at potentials below 3.7 V (Stage I), which is predominantly governed by anionic oxidation. This “anion-first” redox pathway is uniquely enabled by the robust inorganic solid–solid interface, which stabilizes oxidized oxygen species (e.g. O^{•−}). In contrast, in liquid batteries, the initial oxidation of oxygen at low potentials (<3.7 V) readily triggers nucleophilic degradation of carbonate solvents [39], leading to the rapid formation of a resistive cathode-electrolyte interphase (CEI) layer. Therefore, as shown in Figure S15, the intrinsic Stage I plateau is kinetically masked and shifted to a higher potential to overcome the increased interfacial energy barrier, leading the charging process to directly proceed to Stage II (Figure 3b).

To elucidate the fundamental differences in the delithiation behavior of LFO-BM between liquid and solid-state electrolytes, we systematically investigated the coupled ionic–electronic evolution by analyzing Li/O vacancy formation energies, work function, projected density of states (pDOS), and Bader charge (Figures S16–S19, Table S1 and Table S2). The results reveal that, during the initial stage of charging, the liquid electrolyte system is governed predominantly by cationic redox chemistry, whereas the solid-state electrolyte effectively activates lattice oxygen, giving rise to a substantially enhanced anionic redox contribution. These findings further demonstrate that the all-solid-state reaction environment fundamentally reconstructs the redox pathway and charge-compensation mechanism. Detailed computational results and discussions are provided in the Supporting Information.

2.3 | Application of LFO-BM as Prelithiation Additive for Si-Based ASSBs

To evaluate the practical application potential of LFO in Si-based ASSBs, a battery system was constructed using a nickel-rich cathode (LiNi_{0.895}Co_{0.077}Mn_{0.028}O₂, NCM) and a pure Si anode (N/P ratio: 1.1–1.2). A defined amount of LFO-BM (denoted as NCM-F) was incorporated into the composite cathode (NCM, VGCF, and LIC), with the content calculated based on the cathode mass. The Si-based ASSBs with NCM-F cathode exhibit

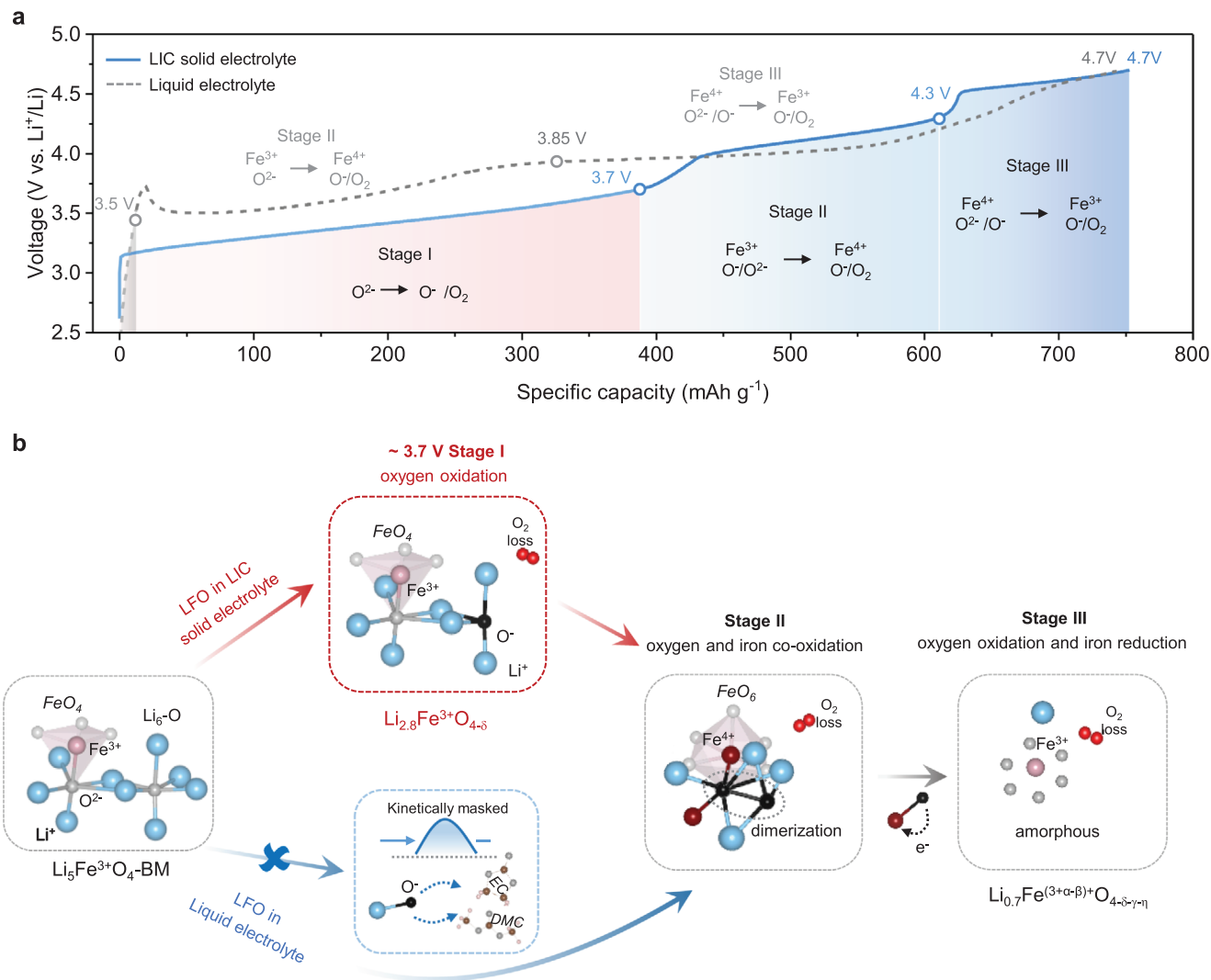


FIGURE 3 | Solid-state-enabled low-potential oxygen redox and three-stage delithiation mechanism of Li_5FeO_4 . (a) Charge curves of LFO-BM. In the solid-state LIC system, a unique three-stage delithiation process is observed.

exceptional rate capability in Figure 4a. At a current density of 2 C, the NCM cathode delivered only 50 mAh g^{-1} , whereas the NCM-F (7%) cathode additive achieved over 100 mAh g^{-1} . Figure 4b,c display the charge–discharge curves of NCM and NCM-F (7%) at 0.1 C. NCM-F exhibits an additional charging plateau around 3.5 V compared to NCM, which corresponds to the delithiation plateau of LFO-BM. Apart from the first charge curve, there is no difference between the other charge–discharge curves of NCM and NCM-F, indicating that the initial delithiation of LFO occurs only during the first charging cycle.

(a) Rate capability of NCM||Si and NCM-F||Si systems within the voltage range of 2.8–4.3 V. (b, c) Initial galvanostatic charge–discharge profiles of (b) NCM||Si and (c) NCM-F(7%)||Si ASSBs at 0.1 C. (d) Long-term cycling stability of NCM||Si and NCM-F(7%)||Si systems at 0.5 C (1 C = 200 mA g^{-1}) under RT. (e) Comparison of energy density and cycling performance between this work and previously reported Si-based battery systems. (f) Low-temperature (-10°C) cycling performance of NCM-F(7%)||Si at 0.2 C. (g) Cycling performance of an all-solid-state pouch cell under a stack pressure of 24.5 MPa, with the inset showing the

corresponding dQ/dV curve and an optical photograph of the device.

The effect of LFO on the cycling stability of Si-based ASSBs has been verified under a wide range of temperature conditions. The cycling performance of these cells was first characterized at room temperature (RT). As demonstrated in Figure 4d, the NCM-F (7%) cathode maintained 81.4% capacity retention after 400 cycles, compared to only 18.1% for the LFO-free NCM cathode, and still retained 70% of its initial capacity after 1000 cycles. In addition, under challenging conditions involving a high mass loading of 21 mg cm^{-2} , the ASSB delivers a stable areal capacity of 4 mAh cm^{-2} retains 60.5% of its capacity after 120 cycles (Figure S20). Rate testing from 0.1 to 0.5 C further demonstrates its rate capability under high-loading conditions. Moreover, even at 60°C and 0.5 C, the high-loading ASSB maintains 80.1% of its capacity after 80 cycles, demonstrating robust cycling stability under combined high-loading and elevated-temperature conditions. Notably, the energy density achieved in this LFO-enhanced Si-based ASSB (calculated based on the cathode mass in the first cycle in Figure 5d) surpasses that of several recently

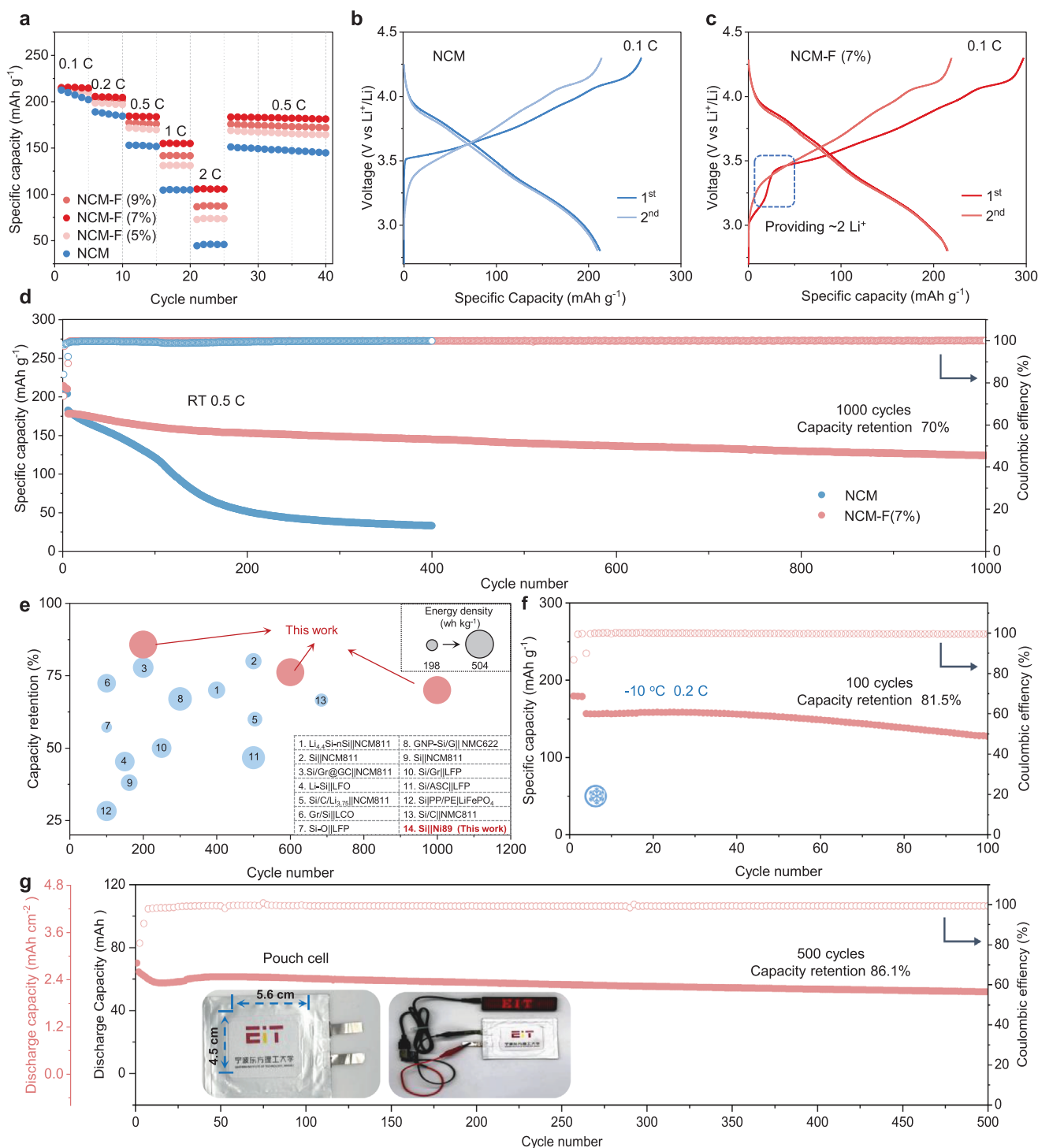


FIGURE 4 | Electrochemical performance of Si-based ASSBs using LFO-BM as a prelithiation additive.

reported Si-based battery systems, particularly when comparing long-term cycling performance, as shown in Figure 4e and Table S3.

Furthermore, at a low temperature of -10°C, the ASSBs with NCM-F cathode achieve a capacity retention of approximately 81.5% after 100 cycles at 0.2 C (Figure 4f). Under a high-temperature condition of 55°C in Figure S21, the ASSBs retain 80% of their initial capacity following 500 cycles at a high rate of 5 C, highlighting the remarkable effectiveness of the prelithiation

agents in extreme environments. Additionally, the all-solid-state pouch cell employing the NCM-F cathode maintains a discharge capacity of approximately 60 mAh and a corresponding energy density of about 473.3 Wh kg⁻¹ (based on electrode), while retaining 86.1% capacity after 500 cycles at 0.33 C and RT under a low pressure of 24.5 MPa in Figure 4g. These results collectively demonstrate that LFO serves as a sacrificial cathode prelithiation additive, thereby significantly enhancing the overall performance of Si-based ASSBs and highlighting its considerable practical potential.

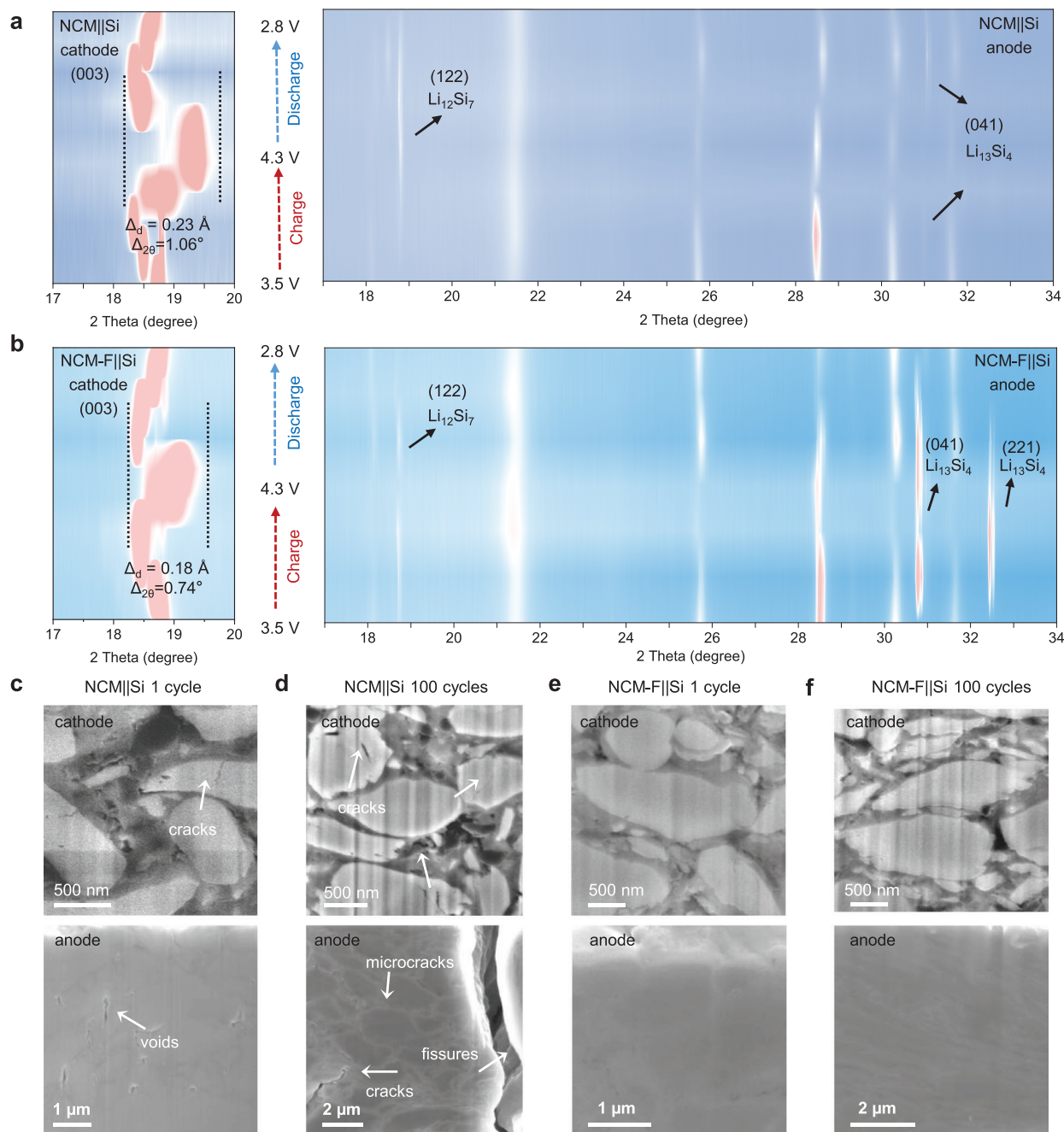


FIGURE 5 | Working mechanism of LFO-BM in ASSB performance enhancement.

2.4 | Dual-Function Mechanism of LFO-BM in ASSB Performance Enhancement

The functionality of LFO-BM, with a focus on its interfacial and structural effects, was investigated through ex situ XRD, SEM, focused ion beam-scanning electron microscope (FIB-SEM), galvanostatic intermittent titration technique (GITT), in situ EIS, and ex situ XPS analysis. In the NCM||Si system, the (003) plane of the NCM cathode undergoes a maximum interplanar spacing change of 0.23 Å during the first cycle (Figure 5a), whereas the NCM-F cathode exhibits a significantly smaller variation of only 0.18 Å (Figure 5b). This observation indicates that LFO-BM effectively mitigates lattice expansion and contraction, thereby

enhancing the structural stability and mechanical integrity of the cathode material [48, 49]. On the anode side, the XRD pattern of the Si electrode in the NCM||Si system reveals only the (122) plane of $\text{Li}_{12}\text{Si}_7$ and the (041) plane of $\text{Li}_{13}\text{Si}_4$. In contrast, the NCM-F||Si system clearly shows diffraction peaks corresponding to three distinct Li-Si alloy phases: the (122) plane of $\text{Li}_{12}\text{Si}_7$, along with the (041) and (221) planes of $\text{Li}_{13}\text{Si}_4$, all with notably higher intensities. These results suggest that LFO-BM promotes in-situ prelithiation of the Si-based anode, facilitating the formation of Li-Si alloys with higher lithium content. Furthermore, regarding the reversibility of the (041) plane of the Li-Si alloy during cycling, the NCM||Si system exhibits poor reversibility, indicating the presence of inactive

lithium (“dead Li”) in the Si anode. In comparison, the NCM-F||Si system demonstrates highly reversible changes in both the (041) and (221) planes, implying that prelithiation contributes to superior electrochemical reversibility.

These structural observations are confirmed by cross-sectional morphology analysis (Figure S22). Upon charging to 3.5 V, the Si anode in the NCM-F||Si system exhibits a greater thickness increase (from 12.5 to 16.8 μm), directly confirming the prelithiation and Li–Si alloy formation facilitated by LFO-BM. The retained thickness and enhanced alloy content after cycling further demonstrate that the lithium ions released from LFO remain in the anode, compensating for irreversible loss and contributing to the improved cycling performance of the system.

To directly observe the internal microstructure and degradation features of the cycled electrodes, focused ion beam-scanning electron microscopy (FIB-SEM) was employed. The corresponding cross-sectional SEM images (Figure 5c–f) reveal a pronounced contrast in the morphological evolution of the cathode and anode between the NCM||Si and NCM-F||Si systems. On the cathode side, NCM particles in the NCM||Si system already show microcracks after the first cycle (Figure 5c), which further propagate and coalesce into extensive cracking after 100 cycles (Figure 5d). In sharp contrast, the NCM cathode in the NCM-F||Si system, stabilized by the LFO-BM additive, maintains excellent mechanical integrity with virtually no detectable cracking even after 100 cycles (Figures 5e and 5f). On the anode side, the pristine Si anode exhibits substantial intrinsic porosity (Figure S23). In the NCM-F||Si system, this porous structure is completely filled after the first cycle, transforming into a dense Li–Si alloy that remains stable throughout 100 cycles (Figure 5e,f). Conversely, in the NCM||Si system, a void-rich structure persists in the anode after the first cycle. Upon extended cycling to 100 cycles, the anode undergoes severe degradation, evolving into microcracks, cracks, and large fissures.

These morphological observations collectively demonstrate that the prelithiation agent LFO-BM enhances the mechanical stability of both electrodes. By suppressing crack initiation and propagation in the cathode and promoting the formation of a dense, stable Li–Si alloy in the anode, it significantly improves interparticle contact and overall structural integrity [50, 51], thereby providing critical support for the superior electrochemical performance of the battery.

(a, b) Ex situ XRD patterns of (a) NCM||Si and (b) NCM-F(7%)||Si systems during the first cycle at 0.1 C. Left panels display the (003) reflection of the NCM cathode, while right panels show the phase transitions of the Si anode. (c–f) Cross-sectional SEM images of the cathode and anode interfaces: (c) NCM||Si after 1 cycle, (d) NCM||Si after 100 cycles, (e) NCM-F(7%)||Si after 1 cycle, and (f) NCM-F(7%)||Si after 100 cycles.

Additionally, the cathode-driven prelithiation strategy, enabled by the LFO-BM additive, delivers synergistic improvements in both kinetics and interfacial stability. GITT and in situ EIS-DRT analyses confirm enhanced Li^+ diffusivity and lower charge-transfer resistance (Figures S24–S27). This kinetic advantage, coupled with a buffered anode state, directly fosters superior

interfacial stability. The latter is evidenced by a stable resistance profile over 100 cycles (Figure S27f) and, at the chemical level, by attenuated electrolyte degradation as shown in ex situ XPS (Figure S28). Collectively, these results demonstrate that the prelithiation strategy effectively optimizes bulk transport and interfacial electrochemistry concurrently.

3 | Conclusion

In summary, we demonstrate that nanosized Li_5FeO_4 serves as an effective cathode prelithiation additive for silicon-based ASSBs. Mechanochemical milling reduces the particle size of Li_5FeO_4 from the micrometer scale to approximately 500 nm, induces structural disorder, changes the local Fe coordination environment, improves electronic conductivity, and increases delithiation capacity of Li_5FeO_4 from 43.1 to 752.5 mAh g^{-1} in halide-based ASSBs. Mechanistic analyses reveal a solid-state-specific three-stage delithiation pathway, involving low-potential lattice oxygen oxidation, coupled oxygen/iron oxidation, and lattice oxygen oxidation accompanied by iron reduction. When used as a cathode prelithiation additive, nanosized Li_5FeO_4 effectively replenishes lithium consumed by silicon anodes, enabling Si-based ASSBs with 70% capacity retention over 1000 cycles and stable operation from -10°C to 55°C . Moreover, a Si-based all-solid-state pouch cell delivers an energy density of 473.3 Wh kg^{-1} based on electrode mass and retains 86.1% capacity after 500 cycles under 24.5 MPa. This work reveals the distinct delithiation chemistry of Li_5FeO_4 in an all-solid-state reaction environment and establishes nanosizing as an effective strategy for transforming Li_5FeO_4 into a high-capacity prelithiation reservoir for high-energy-density ASSBs.

Author Contributions

T.L. and C.W. conceived the project and designed the experimental framework. T.L. carried out the modification of Li_5FeO_4 , materials characterization, and battery performance testing. J.F., J.P., and S.L. collected synchrotron x-ray diffraction data at the Canadian Light Source (CLS) and contributed to data interpretation. M.W. and Z.W. conducted theoretical simulations under the supervision of Z.Z., K.T., M.Z., H.H., and S.X. assisted in the synthesis of SEs. S.L. and K.T. performed TEM characterization. S.X., M.Z., L.X., H.H., and Z.W. contributed to the fabrication and assembly of ASSBs. C.W., Z.W., and S.L. contributed to spectroscopic analyses. C.W. and X.S. supervised the overall research. T.L. and S.X. wrote the original draft with critical input from C.W., Q.Z. and X.S. All authors discussed the results and reviewed the final manuscript.

Acknowledgments

Tingting Liu, Shengjie Xia, and Kaiyong Tuo contributed equally to this work. This work was supported by the National Natural Science Foundation of China (Grant Nos. 92572112, 22409103, W2441017, 52622211, U25A20237, and 92472104), the National Key Research and Development Program of China (2025YFF0523000), the Natural Science Foundation of Zhejiang Province (Grant Number. LZ26E020006), the Commanding Heights of Science and Technology of Chinese Academy of Sciences (LDES150000), the “Innovation Yongjiang 2035” Key R&D Program (Grant Nos. 2025Z063, 2024Z040), the Zhejiang Provincial Natural Science Foundation of China (Grant Nos. LZ26E020006, LMS26E020021, LMS26E020020, Z26E020005), Zhejiang Provincial Key R&D Program of China (Grant Number. 2026C02A1073), the China Postdoctoral Science Foundation (Grant Number. 2025M770126).

Conflicts of Interest

The authors declare no conflicts of interest.

Data Availability Statement

The data that supports the findings of this study are available in the [Supporting Information](#) of this article.

References

1. W. J. Kong, C. Z. Zhao, L. Shen, et al., "Bulk/Interfacial Structure Design of Li-Rich Mn-Based Cathodes for all-Solid-State Lithium Batteries," *Journal of the American Chemical Society* 146 (2024): 28190–28200, <https://doi.org/10.1021/jacs.4c08115>.
2. Y.-G. Lee, S. Fujiki, C. Jung, et al., "High-Energy Long-cycling all-Solid-State Lithium Metal Batteries Enabled by Silver–Carbon Composite Anodes," *Nature Energy* 5 (2020): 299–308, <https://doi.org/10.1038/s41560-020-0575-z>.
3. J. M. Yao, Z. X. Yu, J. Ma, et al., "Revealing Interfacial Failure Mechanism of Silicon Based all Solid State Batteries via Cryogenic Electron Microscopy," *Nature Communications* 16 (2025): 9731, <http://doi.org/10.1038/s41467-025-64697-0>.
4. C. Y. Wang, Y. Q. Jing, D. Zhu, and H. L. Xin, "Atomic Origin of Chemomechanical Failure of Layered Cathodes in all-Solid-State Batteries," *Journal of the American Chemical Society* 146 (2024): 17712–17718, <https://doi.org/10.1021/jacs.4c02198>.
5. J. M. Fu, C. H. Wang, S. Wang, et al., "A Cost-Effective all-in-one Halide Material for all-Solid-State Batteries," *Nature* 643 (2025): 111–118, <https://doi.org/10.1038/s41586-025-09153-1>.
6. K. Tuo, C. W. Sun, and S. Q. Liu, "Recent Progress in and Perspectives on Emerging Halide Superionic Conductors for all-Solid-State Batteries," *Electrochemical Energy Reviews* 6 (2023): 17, <https://doi.org/10.1007/s41918-023-00179-5>.
7. M. X. Zuo, X. Hu, C. Z. Ji, et al., "Gradient-Interlocked Solid Electrolyte Interphase via Spatial Reconfiguration for Stable Silicon Anodes in Solid-State Batteries," *Journal of the American Chemical Society* 148 (2026): 5222–5231, <https://doi.org/10.1021/jacs.5c17488>.
8. J. Janek and W. G. Zeier, "Challenges in Speeding up Solid-state Battery Development," *Nature Energy* 8 (2023): 230–240, <https://doi.org/10.1038/s41560-023-01208-9>.
9. Y. Jiang, S. Yao, H. Feng, et al., "Staged Lithiation/Delithiation of Silicon Anode in all-Solid-State Batteries Revealed by High-Stack-Pressure Operando NMR Spectroscopy," *Journal of the American Chemical Society* 148 (2026): 15024–15034, <https://doi.org/10.1021/jacs.5c23072>.
10. Z. Wu, X. Li, C. Zheng, et al., "Interfaces in Sulfide Solid Electrolyte-Based all-Solid-State Lithium Batteries: Characterization, Mechanism and Strategy," *Electrochemical Energy Reviews* 6 (2023): 10, <https://doi.org/10.1007/s41918-022-00176-0>.
11. S. Sun, C. Z. Zhao, H. Yuan, et al., "Eliminating Interfacial O-involving Degradation in Li-rich Mn-based Cathodes for all-solid-state Lithium Batteries," *Science Advances* 8 (2022): 44.
12. G. X. Liu, W. Wan, Q. Nie, et al., "Controllable Long-term Lithium Replenishment for Enhancing Energy Density and Cycle Life of Lithium-ion Batteries," *Energy & Environmental Science* 17 (2024): 1163–1174, <https://doi.org/10.1039/D3EE03740A>.
13. W. Zhong, R. J. He, L. F. Peng, et al., "Lifecycle Synergistic Prelithiation Strategy of Both Anode and Cathode for High-Performance Lithium-Ion Batteries," *Advanced Energy Materials* 15 (2025): 202406007, <https://doi.org/10.1002/aenm.202406007>.
14. S. Lou, F. Zhang, C. Fu, et al., "Interface Issues and Challenges in All-Solid-State Batteries: Lithium, Sodium, and Beyond," *Advanced Materials* 33 (2021): e2000721, <https://doi.org/10.1002/adma.202000721>.
15. L. Quan, Q. L. Su, H. Z. Lei, et al., "Integrated Prelithiation and SEI Engineering for High-Performance Silicon Anodes in Lithium-ion Batteries," *National Science Review* 12 (2025): 7, <https://doi.org/10.1093/nsr/nwaf084>.
16. W. L. Yan, Z. L. Mu, Z. X. Wang, et al., "Hard-Carbon-Stabilized Li–Si Anodes for High-Performance All-Solid-State Li-ion Batteries," *Nature Energy* 8 (2023): 800–813, <https://doi.org/10.1038/s41560-023-01279-8>.
17. Z. Zhang, Z. Sun, X. Han, et al., "An All-electrochem-active Silicon Anode Enabled by Spontaneous Li–Si Alloying for Ultra-High Performance Solid-State Batteries," *Energy & Environmental Science* 17 (2024): 1061–1072, <https://doi.org/10.1039/D3EE03877G>.
18. C. C. Wang, W. J. Jeong, D. L. Nelson, H. Sridhara, S. N. Anderson, and M. T. McDowell, "Prelithiation of Alloy Anodes via Roll Pressing for Solid-State Batteries," *Advanced Materials* 37 (2025): e08973, <https://doi.org/10.1002/adma.202508973>.
19. Y. Ji, Z. Wang, C. Zhao, et al., "Diffusion-Free all-Solid-State Batteries Enabled by an Ionic/Electronic Dual-Conductive Anode," *Renewables* 2 (2024): 194–203, <https://doi.org/10.31635/renewables.024.202400056>.
20. Y. L. Zhu, Y. L. Chen, J. K. Chen, et al., "Lattice Engineering on Li₂CO₃-Based Sacrificial Cathode Prelithiation Agent for Improving the Energy Density of Li-Ion Battery Full-Cell," *Advanced Materials* 36 (2024): 2312159, <https://doi.org/10.1002/adma.202312159>.
21. L. Chen, Y. L. Zhu, W. H. Zuo, et al., "Implanting Transition Metal Into Li₂O-Based Cathode Prelithiation Agent for High-Energy-Density and Long-Life Li-Ion Batteries," *Angewandte Chemie International Edition* 63 (2024): e202316112, <https://doi.org/10.1002/anie.202316112>.
22. M. Fan, Q. H. Meng, X. Chang, et al., "In Situ Electrochemical Regeneration of Degraded LiFePO₄ Electrode With Functionalized Prelithiation Separator," *Advanced Energy Materials* 12 (2022): 2103630, <https://doi.org/10.1002/aenm.202103630>.
23. T. Lv, W. Zhong, R. J. He, et al., "Low-potential Li₂C₂O₄ Prelithiation Catalyzed by 2D MoN With Dominant (002) Crystal Face for High-energy Lithium-ion Batteries," *Materials Science and Engineering R: Reports* 165 (2025): 101014, <https://doi.org/10.1016/j.mser.2025.101014>.
24. G. X. Huang, J. N. Liang, X. G. Zhong, et al., "Boosting the Capability of Li₂C₂O₄ as Cathode Pre-lithiation Additive for Lithium-ion Batteries," *Nano Research* 16 (2023): 3872–3878, <https://doi.org/10.1007/s12274-022-5146-0>.
25. Y. M. Sun, H.-W. Lee, W. Z. Seh, et al., "High-capacity Battery Cathode Prelithiation to Offset Initial Lithium Loss," *Nature Energy* 1 (2016): 15008, <https://doi.org/10.1038/nenergy.2015.8>.
26. M. Diaz-Lopez, P. A. Chater, P. Bordet, et al., "Li₂O:Li–Mn–O Disordered Rock-Salt Nanocomposites as Cathode Prelithiation Additives for High-Energy Density Li-Ion Batteries," *Advanced Energy Materials* 10 (2020): 1902788, <https://doi.org/10.1002/aenm.201902788>.
27. L. Y. Zheng, A. S. Yu, G. Li, and J. J. Zhang, "High-Energy-Density and Long-Lifetime Lithium-Ion Battery Enabled by a Stabilized Li₂O₂ Cathode Prelithiation Additive," *ACS Applied Materials & Interfaces* 14 (2022): 38706–38716, <https://doi.org/10.1021/acsami.2c08788>.
28. J. Y. Shi, C. C. Su, R. Amine, et al., "Prelithiation of Lithium Peroxide for Silicon Anode: Achieving a High Activation Rate," *ACS Appl Mater Interfaces* 15 (2023): 26710–26717, <https://doi.org/10.1021/acsami.3c03312>.
29. K. J. Jun, L. Kaufman, W. M. Jung, et al., "Understanding the Irreversible Reaction Pathway of the Sacrificial Cathode Additive Li₆CoO₄," *Advanced Energy Materials* 13 (2023): 2301132, <https://doi.org/10.1002/aenm.202301132>.
30. H. C. Wang, Y. T. Liu, M. Z. Jiang, et al., "The Potential Regulation of Working Anode for Long-Term Zero-Volt Storage at 37°C in Li-Ion Batteries," *Advanced Materials* 36, no. 25 (2024): 2400656, <https://doi.org/10.1002/adma.202400656>.
31. M. Y. Tian, Y. Yan, H. L. Yu, et al., "Designer Lithium Reservoirs for Ultralong Life Lithium Batteries for Grid Storage," *Advanced Materials* 36 (2024): 2400707, <https://doi.org/10.1002/adma.202400707>.
32. C. S. Liu, H. Zhang, W. W. Zhou, et al., "Air-Stable Li₅FeO₄ Additive Enabled by Carbon Coating for Energy-Dense Lithium-ion Batteries,"

Nature Communications 16 (2025): 7694, <https://doi.org/10.1038/s41467-025-62418-1>.

33. C. Zhan, Z. Yao, J. Lu, et al., “Enabling the High Capacity of Lithium-rich Anti-fluorite Lithium Iron Oxide by Simultaneous Anionic and Cationic Redox,” *Nat Energy* 2 (2017): 963–971, <https://doi.org/10.1038/s41560-017-0043-6>.

34. H. Kobayashi, Y. Nakamura, M. Nakayama, S. Kodaki, R. Matsuo, and I. Honma, “Metastable Cubic Structure Exceeds Capacity Limit of Antifluorite Li_5FeO_4 Cathode Using Small Polarized Oxygen Redox,” *Advanced Energy Materials* 13 (2023): 2203441, <https://doi.org/10.1002/aenm.202203441>.

35. C. S. Johnson, S. H. Kang, J. T. Vaughey, S. V. Pol, M. Balasubramanian, and M. M. Thackeray, “ Li_2O Removal From Li_5FeO_4 : A Cathode Precursor for Lithium-Ion Batteries,” *Chemistry of Materials* 22 (2010): 1263–1270, <https://doi.org/10.1021/cm902713m>.

36. T. Okumura, M. Shikano, and H. J. J. O. M. C. A. Kobayashi, “Effect of Bulk and Surface Structural Changes in Li_5FeO_4 Positive Electrodes During First Charging on Subsequent Lithium-ion Battery Performance,” *Journal of Materials Chemistry A* 2 (2014): 11847–11856, <https://doi.org/10.1039/C4TA01884B>.

37. X. Q. Xu, S. Y. Chu, S. Xu, et al., “Integrating Prelithiation and Interface Protection to Achieve High-Energy All-Solid-State Batteries,” *Angewandte Chemie International Edition* 64 (2025): e202415891, <https://doi.org/10.1002/anie.202415891>.

38. C.-F. Gu, C. Li, X. Chang, et al., “Multiphase Nanocomposite Prelithiation Agents for Service Life Extension of Silicon-Based Lithium-Ion Batteries,” *Science Advances* 12 (2026): ead9626.

39. Y. L. Zhu, R. Y. Xu, Y. C. Zheng, et al., “Unravelling the Oxygen Evolution Mechanism of Lithium-Rich Antifluorite Prelithiation Agent Based on Anionic Oxidation,” *Angewandte Chemie International Edition* 64, no. 19 (2025): e202502126, <https://doi.org/10.1002/anie.202502126>.

40. Q. Meng, M. Fan, X. Chang, et al., “A Functional Prelithiation Separator Promises Sustainable High-Energy Lithium-Ion Batteries,” *Advanced Energy Materials* 13 (2023): 2300507, <https://doi.org/10.1002/aenm.202300507>.

41. R. V. Thøgersen, H. H. Hval, and H. Fjellvåg, “Elucidation of the Reaction Mechanisms in Antifluorite-Type $\text{Li}_{5+x}\text{Fe}_{1-x}\text{Co}_x\text{O}_4$ Positive Electrodes for Li-Ion Batteries,” *Batteries Supercaps* 7 (2024): e202400348, <https://doi.org/10.1002/batt.202400348>.

42. R. A. House, U. Maitra, M. A. Pérez-Osorio, et al., “Superstructure Control of First-Cycle Voltage Hysteresis in Oxygen-redox Cathodes,” *Nature* 577 (2020): 502–508, <https://doi.org/10.1038/s41586-019-1854-3>.

43. H. Kobayashi, T. Makimoto, Y. Ogasawara, et al., “Universal Solid-state Oxygen Redox in Antifluorite Lithium Oxides via Transition Metal Doping,” *Materials Advances* 1 (2020): 1301–1306, <https://doi.org/10.1039/D0MA00395F>.

44. J. Auvergniot, A. Cassel, J. B. Ledeuil, V. Viallet, V. Seznec, and R. Dedryvère, “Interface Stability of Argyrodite $\text{Li}_6\text{PS}_5\text{Cl}$ Toward LiCoO_2 , $\text{LiNi}_{1/3}\text{Co}_{1/3}\text{Mn}_{1/3}\text{O}_2$, and LiMn_2O_4 in Bulk All-Solid-State Batteries,” *Chemistry of Materials* 29 (2017): 3883–3890, <https://doi.org/10.1021/acs.chemmater.6b04990>.

45. S. Wang C. Lou, X. Wu, J. Lin, et al., “Large-Scale Manufacturing Sulfide Superionic Conductor for Advancing All-Solid-State Batteries,” *Matter* 8 (2025): 102135, <http://doi.org/10.1016/j.matt.2025.102135>.

46. F. Walther, R. Koerver, T. Fuchs, et al., “Visualization of the Interfacial Decomposition of Composite Cathodes in Argyrodite-Based all-Solid-State Batteries Using Time-of-Flight Secondary-Ion Mass Spectrometry,” *Chemistry of Materials* 31 (2019): 3745–3755, <https://doi.org/10.1021/acs.chemmater.9b00770>.

47. Z. P. Yao, M. K. Y. Chan, and C. Wolverton, “Exploring the Origin of Anionic Redox Activity in Super Li-Rich Iron Oxide-Based High-Energy-Density Cathode Materials,” *Chemistry of Materials* 34 (2022): 4536–4547, <https://doi.org/10.1021/acs.chemmater.2c00322>.

48. T. T. Liu, J. Ma, T. He, et al., “Practical Surface-Reconstruction Strategy Based on Self-Reactive Interface Design Enables Ultralong Cyclability of Commercial Cathodes for Lithium-ion Batteries,” *Energy Storage Materials* 71 (2024): 103577, <https://doi.org/10.1016/j.ensm.2024.103577>.

49. X. H. Tan, Y. X. Zhang, S. Y. Xu, et al., “High-Entropy Surface Complex Stabilized LiCoO_2 Cathode,” *Advanced Energy Materials* 13 (2023): 2300147, <https://doi.org/10.1002/aenm.202300147>.

50. C. Y. Lu, M. Chao, Y. C. Wu, et al., “Robust 3D Carbon Host Decorated With in-Situ Generated Li-Si Alloy and Li_3N for Li Metal Anodes,” *Carbon* 225 (2024): 119146, <https://doi.org/10.1016/j.carbon.2024.119146>.

51. D. H. Kim, Y. H. Lee, J. M. Yoon, et al., “Li-Si Compound Anodes Enabling High-Performance all-Solid-State Li-ion Batteries,” *Joule* 9 (2025): 102195, <https://doi.org/10.1016/j.joule.2025.102195>.

Supporting Information

Additional supporting information can be found online in the Supporting Information section.

Supporting File: anie74878-sup-0001-SuppMat.docx.